



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
REVIEW TEST
10-10-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TEAS)

Time : 3 Hours

PAPER-2

Maximum Marks : 216

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

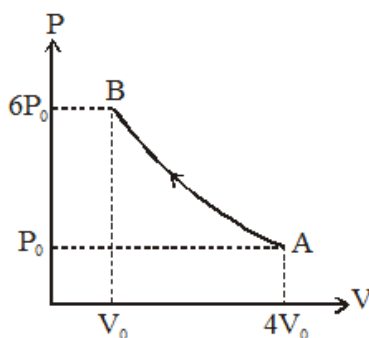
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

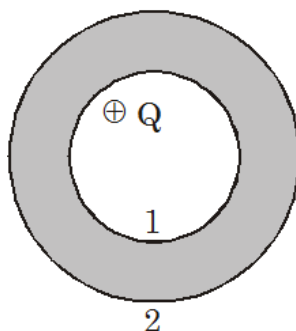
1. One mole of a diatomic gas is taken through a process $A \rightarrow B$ as shown in the figure. The gas obeys the relation $Q_{A \rightarrow B} + W_{A \rightarrow B} = 0$



- (A) The molar heat capacity for process AB is $5R/4$
 (B) The heat supplied in the process is $(5/2) P_0 V_0$
 (C) For A to B, temperature initially decreases and then increases
 (D) The temperature goes on increasing from A to B

Space for Rough Work

2. The positive charge Q is placed in the cavity of the spherical conductor.



- (A) the induced charge at the surface 1 is $-Q$ and on the surface 2 is $+Q$
- (B) the surface charge density at inner surface is uniform
- (C) the surface charge density at the outer surface is uniform
- (D) the charge Q does not experience electric force if it is placed at the centre of the spherical cavity.
3. A wave disturbance in a medium is described by $y(x, t) = 0.02 \cos(50\pi t + \pi/2) \cos(10\pi x)$, where x and y are in metres and t in seconds.
- (A) A node occurs at $x = 0.15$ m
- (B) An antinode occurs at $x = 0.3$ m
- (C) The speed of the component wave is 5.0 m/s
- (D) The wavelength is 0.2 m

Space for Rough Work

4. A closed vessel contains a mixture of two diatomic gases A and B. Molar mass of A is 16 times that of B and mass of gas A, contained in the vessel is 2 times that of B.
- (A) Average kinetic energy per molecule of gas A is equal to that of gas B
- (B) Root mean square value of translational velocity of gas B is four times that of A
- (C) Pressure exerted by gas B is eight times of that exerted by gas A
- (D) Number of molecules of gas B in the cylinder is eight times that of gas A
5. For a single material
- (A) its absorptive power and emissivity, have equal magnitude
- (B) molar specific heat capacity is proportional to its specific heat capacity.
- (C) mass is proportional to number of moles.
- (D) density is proportional to volume of material.
6. A black body has a temperature of 'T' (in kelvin) and wavelength ' λ ' corresponding to maximum energy density. When body cools, percentage change in the wavelength corresponding to the maximum energy density is 900% Then choose the correct alternative(s)
- (A) Temperature of black body to which it is cooled is $\frac{T}{9}$
- (B) Ratio of rate of emission i.e. final to initial will be 1 : 10000
- (C) The magnitude of difference in two temperature is $\frac{9T}{10}$
- (D) Ratio of rate of emission i.e. final to initial will be 1 : 6561

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 16)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has Four options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

Paragraph for Questions 7 and 8

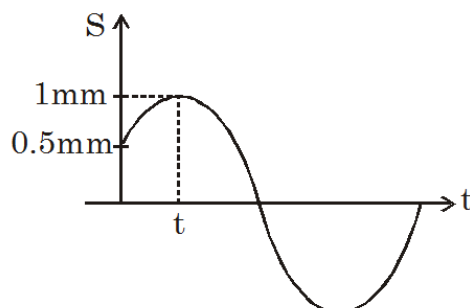
A universally accepted model of nucleus is that it is a spherical volume with uniform charge distribution. Let us consider a ${}_{92}^{238}\text{U}$ atom which is stripped of all its electrons. The radius of nucleus is 10^{-15} m. Consider an electron on the surface of the nucleus. Take : $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$

7. Choose the correct statement(s) :
 (A) The potential energy of interaction of electron and nucleus is of order of 10^8 eV .
 (B) The self energy of nucleus is of order of 10^8 eV
 (C) The attractive force experienced by e^- is of order of 10^4 N
 (D) The potential energy of interaction is greater in modulus than self energy of nucleus.
8. We consider a nucleus of charge Ze (assume to be fixed). The electron is to be projected from the surface so that it escapes the influence of the nucleus. What can be value of Z & r so that e^- cannot escape nucleus. i.e., escape speed required is greater than speed of light. $m_e = 9 \times 10^{-31} \text{ kg}$. This is an example of electrostatic black hole. Assume classical physics.
 (A) $Z = 2, r_0 = 10^{-14} \text{ m}$ (B) $Z = 40, r_0 = 2 \times 10^{-13} \text{ m}$
 (C) $Z = 35, r_0 = 2 \times 10^{-13} \text{ m}$ (D) $Z = 20, r_0 = 1.6 \times 10^{-13} \text{ m}$

Space for Rough Work

Paragraph for Questions 9 and 10

A long copper rod has one end at $x = 0$. This end is subjected to longitudinal oscillations by an external agent with amplitude of 1mm and frequency of 400 Hz. This produces longitudinal travelling waves through the medium. Assume that the medium is non dispersive & non dissipative. The Young's modulus of elasticity for copper is 10^{11} N/m^2 and density is 7840 kg/m^3 , Area of rod is 10 cm^2 . The displacement of end at $x = 0$ v/s time graph is shown here. The wave travels in +ve x direction.



9. Select the correct statement(s) :

(A) $t = \frac{1}{2400} \text{ sec}$

(B) Force exerted by external agent at $x = 0$ at $t = 0$ is in +ve direction.

(C) Force exerted by external agent at $x = 0$ at $t = 0$ is in -ve direction.

(D) The wavelength is $\frac{125}{14} \text{ m}$.

10. Consider a point P on the rod. At the instant it's velocity is maximum and in +ve direction.

(A) The element is in compression.

(B) It's elastic potential energy is zero.

(C) It's elastic potential energy is maximum.

(D) Maximum velocity is $\frac{4\pi}{5} \text{ m/s}$.

Space for Rough Work

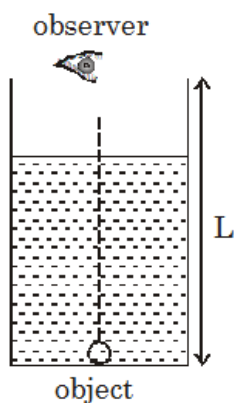
SECTION-II : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

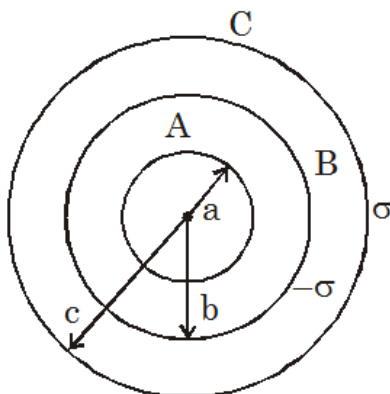
1. A cylinder is filled with a liquid of refractive index μ . The radius of the cylinder is decreasing at a constant rate K . The volume of the liquid inside the container remains constant at V . The observer and the object O are in a state of rest and at a distance L from each other. The apparent speed (in m/s) of the object as seen by the observer, (when radius of cylinder is r)



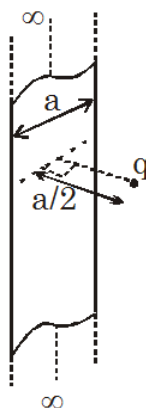
Take $r = 10$ cm, $\mu = 4/3$, $V = \pi \text{ m}^3$, $K = 1$ cm/sec.

Space for Rough Work

2. The adjacent figure shows charged spherical shells A, B and C having charge densities σ , $-\sigma$, σ and radii a , b , c respectively. If $V_A = V_C$ then c is equal to (in m) (assuming $a = 0.10$ m, $b = 0.20$ m)



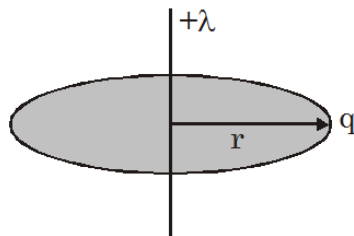
3. There is an infinite long straight surface having width 'a'. A point charge q is placed at a perpendicular distance $\frac{a}{2}$ from the surface symmetrically as shown in the figure. If the flux linked with this infinitely long surface due to charge q is $\frac{q}{k\epsilon_0}$, then find the value of k is



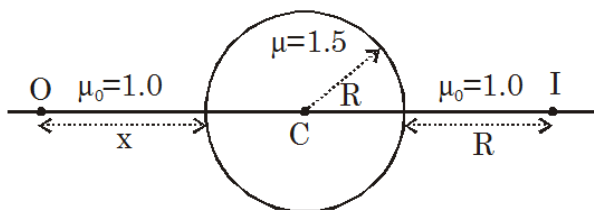
4. An organ pipe P_1 closed at one end vibrating in its first overtone and another pipe P_2 open at the both ends vibrating in its third overtone are in resonance with a given tuning fork. The ratio of the length of P_1 to that of P_2 is $\frac{3}{n}$ then $n = ?$

Space for Rough Work

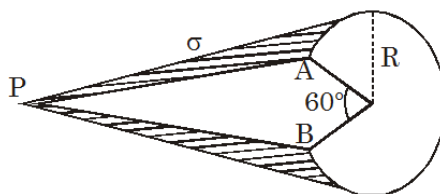
5. If the sound heard by observer, whose equation is given as $y = 8 \sin 10 \pi t \cos 200 \pi t$ at $x = 0$, what is the beat frequency (in Hz) heard by the observer ?
6. A particle of charge $-q$ and mass m moves in a circle of radius r around an infinitely long charge of linear charge density $+\lambda$. Then time period will be dependent on radius r as r^n . Find n .



7. A glass sphere of refractive index 1.5 forms the real image of object O at point I as shown in the figure when kept in air of refractive index 1. If the value of x is kR , find k .



8. A non-conducting hollow cone has charge density σ . A part ABP is cut and removed from the cone. The potential due to the remaining portion of the cone at point 'P' is $c \frac{\sigma R}{\epsilon_0}$. What is the value of c ?



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. For 'n' moles of a van der waal gas undergoes a reversible isothermal process from P_1, V_1, T to P_2, V_2, T . Select the correct statements(s) -

(A) $W = -nRT \ln \left(\frac{V_2 - nb}{V_1 - nb} \right) + an^2 \left(\frac{1}{V_2} - \frac{1}{V_1} \right)$

(B) $\Delta U = -an^2 \left(\frac{1}{V_2} - \frac{1}{V_1} \right)$

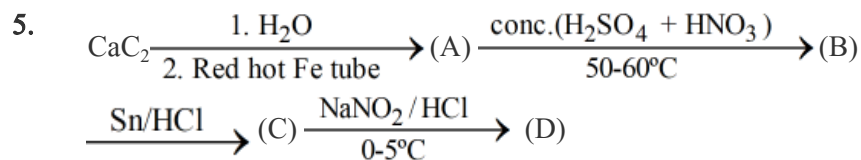
(C) $q = nRT \ln \left(\frac{V_2 - nb}{V_1 - nb} \right)$

(D) $\Delta H = -2an^2 \left(\frac{1}{V_2} - \frac{1}{V_1} \right) + n^2b RT \left(\frac{1}{V_2 - nb} - \frac{1}{V_1 - nb} \right)$

Space for Rough Work

2. EMF of the cell: $\text{Cd(s)} \mid \text{CdCl}_2 \cdot 5\text{H}_2\text{O (sat.)} \mid \text{AgCl(s)} \mid \text{Ag(s)}$ is +0.70 V at 0°C and +0.60 V at 50°C . If ΔH° and ΔS° are temperature independent, then the correct information(s) regarding the cell reaction is/are
- (A) $\Delta G^\circ = -115.8 \text{ kJ at } 50^\circ\text{C}$ (B) $\Delta G^\circ = -135.1 \text{ kJ at } 0^\circ\text{C}$
 (C) $\Delta S^\circ = -386 \text{ J/K}$ (D) $\Delta H^\circ = -221.178 \text{ kJ}$
3. Which of the following statements is/are **CORRECT** for K_2MnO_4 ?
- (A) It is prepared by MnO_2 in acidic medium
 (B) It produce brown coloured compound on addition of water
 (C) It produce KMnO_4 on addition of water
 (D) It produce purple colour compound with O_3
4. When crystals of potassium chloride is heated with Mn_3O_4 (mixed oxide) and conc. H_2SO_4 then :-
- (A) Cl_2 is produced
 (B) Its redox change
 (C) Resulting mixture after dilution by water gives buff colour ppt. with H_2S
 (D) Resulting mixture after dilution by water gives ppt. with excess. KOH

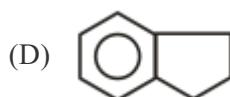
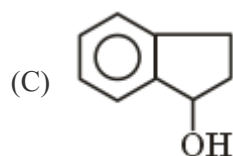
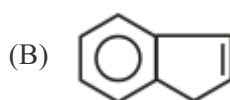
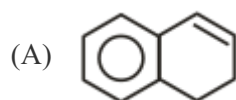
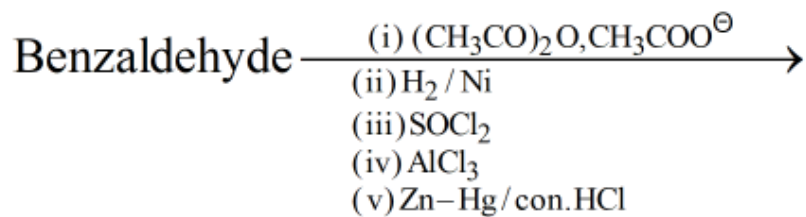
Space for Rough Work



Select correct statement among following for above sequence of reaction :

- (A) 'D' is phenol
 (B) 'A' is an aromatic compound
 (C) 'C' is primary 1° amine
 (D) To convert B to C Fe/HCl can also be used

6. Major product of reaction sequence is :



Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 16)

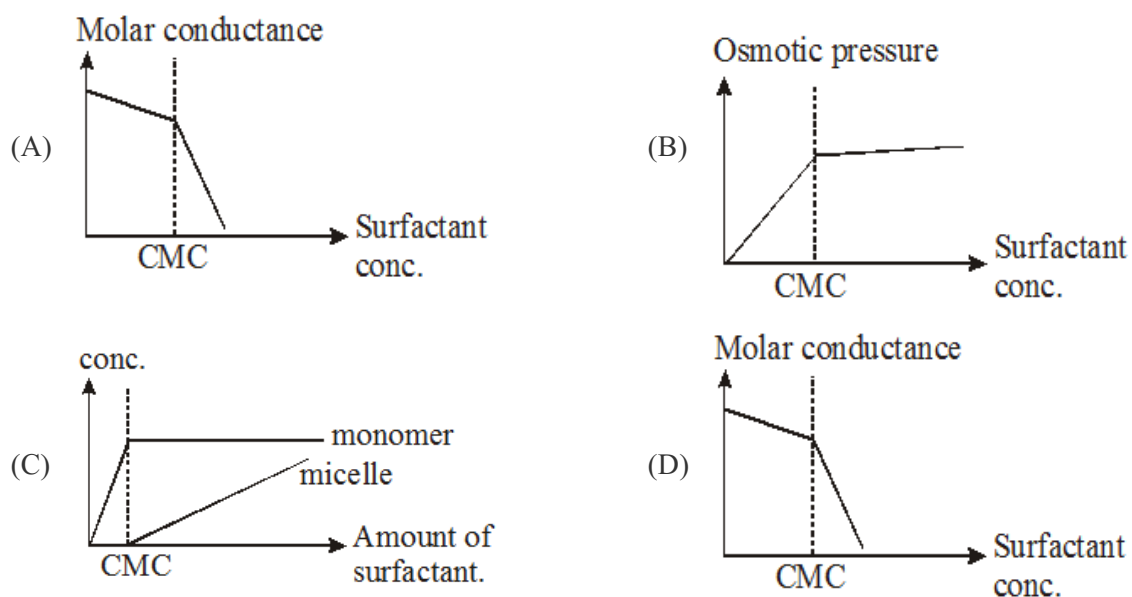
- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has Four options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

Paragraph for Questions 7 and 8

The properties of surfactant at low concentration in water are similar to those of simple electrolytes. At a certain concentration surfactants monomer assemble to form micelle in which the hydrophobic tails are shielded from water while the hydrophilic heads face water, this concentration is called as critical micelle concentration (CMC). CMC is the property of surfactant. Above CMC concentration of monomers are almost constant.

Space for Rough Work

7. Which of the following is not correct -



8. Which of the following is correct -

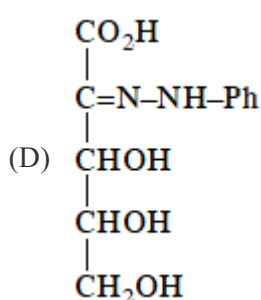
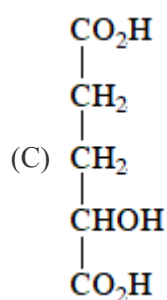
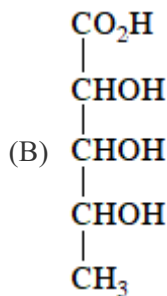
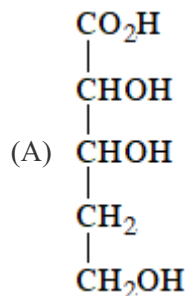
- (A) Micelle formation takes place below a certain temperature called kraft temperature
- (B) Sodium stearate removes oils & fats by emulsification
- (C) Surfactants with big hydrophobic part has higher CMC
- (D) Surfactants must be ionic in nature.

Space for Rough Work

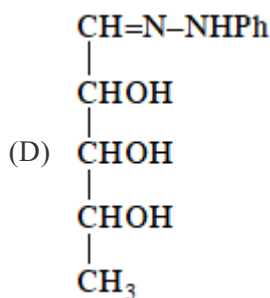
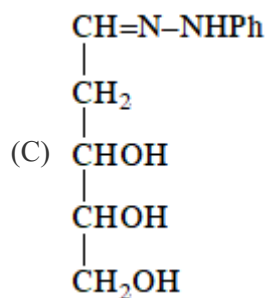
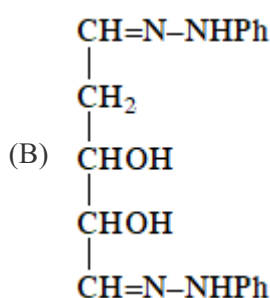
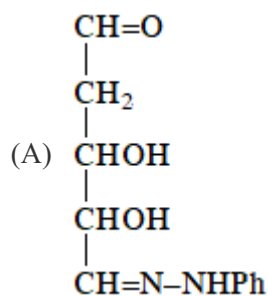
Paragraph for Questions 9 and 10

Compound A ($C_5H_{10}O_4$) on oxidation with $Br_2 + H_2O$ gives acid B ($C_5H_{10}O_5$). A forms triacetate by acetic anhydride and is reduced by red P + HI to give n-pentane. Oxidation of A with HIO_4 gives among other products, 1 mole of $H_2C=O$ & 1 mole of HCO_2H .

9. The possible structure of B is :



10. A reacts with phenyl Hydrazin ($PhNH-NH_2$) to form :



Space for Rough Work

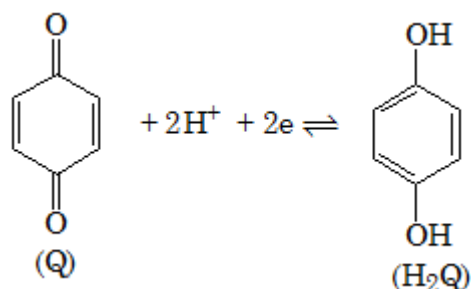
SECTION-II : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. ^{96}Mo crystallizes in bcc structure with a unit cell dimensions of 0.4 nm, its specific gravity is –
[$N_A = 6 \times 10^{23}$]
2. Quinhydrone is a sparingly soluble one-to-one addition compound formed from hydroquinone and quinone. When solid quinhydrone is dissolved in an aqueous medium, equal concentrations of hydroquinone and quinone result. If a platinum wire is dipped into the solution, the potential of the electrode is governed by the reversible reaction :



Where Q denotes quinone and H_2Q denotes hydroquinone. The E° value of this half cell reaction is +0.46V with respect to the saturated calomel reference electrode (SCE). Because the potential of the quinone-hydroquinone half reaction depends on the concentration of hydrogen ion, it is possible to use this system to measure pH. A sample solution of unknown pH was saturated with quinohydrone and a platinum electrode was dipped into it. If the potential of the such electrode was found to be +0.22V with respect to SCE, what was the PH of the sample solution?

[Given : $\frac{2.303 RT}{F} = 0.06 \text{ V}$]

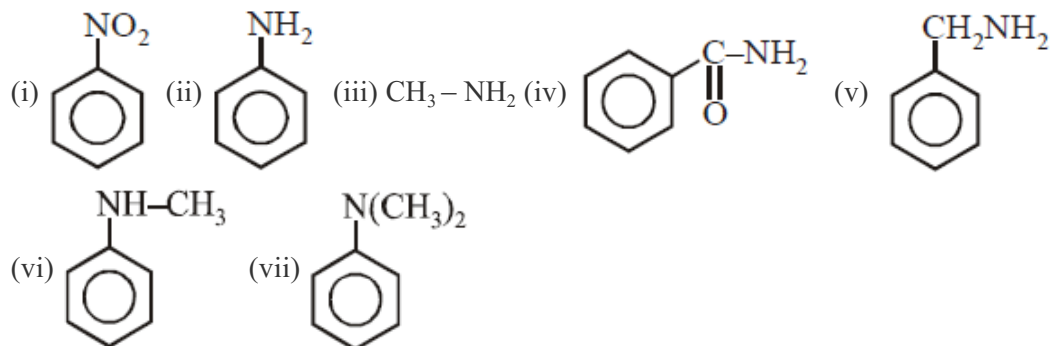
Space for Rough Work

3. Total number of true statements about silicones/silicates ?
- Silicones resist oxidation and attack by organic reagents
 - Silicones are lubricants at low as well as high temperature
 - Cross-linked silicones are formed by the hydrolysis of trichlorosilicone (RSiCl_3) and subsequent polymerisation
 - Both silicones and silicates compound contains Si–O–Si linkage.
 - Silicones are good electrical insulator and are antifoaming agents
4. Consider following reactions :
- $\text{NaH} + \text{H}_2\text{O} \rightarrow$
 - $\text{H}_2\text{S} + \text{Cl}_2 \text{ water} \rightarrow$
 - $\text{AgNO}_3 + \text{H}_2\text{O} + \text{H}_3\text{PO}_2 \rightarrow$
 - $(\text{NH}_4)_2\text{Cr}_2\text{O}_7(\text{s}) \xrightarrow{\Delta}$
 - $\text{H}_2\text{S} + \text{SO}_{2(\text{aq.})} \rightarrow$
 - $\text{XeF}_2 + \text{H}_2\text{O} \rightarrow$
- In how many of the reaction(s) at least one of the product is obtained in elemental state ?
5. Total number of methods which can remove permanent hardness of water.
- Clark's method
 - Ion-exchange method
 - Synthetic resin method
 - Calgon method
 - Treatment with sodium carbonate
6. Select reaction(s) which produce XO_2 type gas (X : central atom, O = oxygen)
- Dolomite $\xrightarrow{\text{dil. HNO}_3}$
 - Dolomite $\xrightarrow{\text{Calcination}}$
 - Zinc blend $\xrightarrow{\text{Roasting}}$
 - Zinc $\xrightarrow{\text{dil. HNO}_3}$
 - $\text{K}_2\text{SO}_3 \xrightarrow{\text{dil. H}_2\text{SO}_4}$
 - Sulphur $\xrightarrow[\text{H}_2\text{SO}_4]{\text{conc.}}$

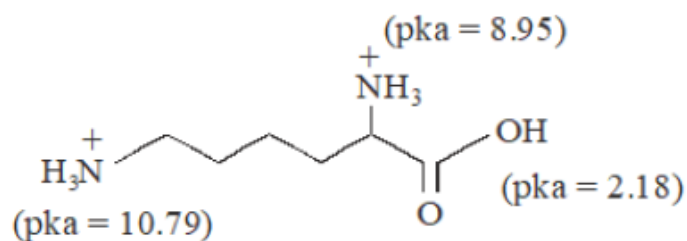
Space for Rough Work

7. x is number of organic compounds soluble in aq. HCl
 y is number of organic compounds which give foul smell with CHCl_3 and alcoholic KOH.
 z is number of organic compounds which reacts with $\text{NaNO}_2 + \text{HCl}$ ($0-5^\circ\text{C}$) followed by β -naphthol gives coloured dye.

Then value of $(x + y - z)$ is :



8. What is the P^I (Isoelectric point) of the following amino acids ?



Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. If $|\vec{a}| = |\vec{b}| = |\vec{c}| = |\vec{d}| = 1$ such that $(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) = 1$ and $\vec{a} \cdot \vec{c} = \frac{1}{2}$, then which of the following can be correct ?

- (A) $\vec{a} \cdot \vec{b} = 0$ (B) $\vec{a} \cdot (\vec{b} \times \vec{c}) = 0$ (C) $\vec{a} \cdot (\vec{b} \times \vec{c}) \neq 0$ (D) $|\vec{a} + \vec{b} + \vec{c}| = \sqrt{4 + \sqrt{3}}$

Space for Rough Work

2. Let $P_1 \equiv x + y + z = 6$ & $P_2 \equiv x + y + z = 11$ be the two parallel planes. Let three points A, B & C are such that their x, y & z co-ordinates can take only positive integral values, then -
- (A) if point A & origin lies on same side of plane P_1 , then A can take 10 different locations in space.
- (B) if point B & origin lies on same side of plane P_2 , then B can take 10 different locations in space.
- (C) if point B & origin lies on same side of plane P_2 , then B can take 120 different locations in space.
- (D) if point C lies between planes P_1 & P_2 , then it can take 100 different locations in space.
3. OABC is a tetrahedron of volume $\frac{2}{3}$ cubic unit. Also $BC = \sqrt{2}$, $\angle AOB = \frac{\pi}{4}$ and $OA + OB = 4$, then
- (A) area of triangle AOB cannot exceed $\sqrt{2}$ sq. unit
- (B) the length of altitude drawn from vertex C to the opposite base is $\sqrt{2}$ units.
- (C) $OC = 2$ unit
- (D) $AC > (2\sqrt{2} - 1)$
4. A and B are non singular square matrices of same order with real entries such that $\text{tr}(AA^T - AB^T A^T) = \text{tr}(ABA^T - ABB^T A^T)$, then which of the following is/are true :
- (A) $\text{tr}(A - AB) \neq 0$
- (B) $A^T = B^T A^T$
- (C) Sum of all element of matrix $A - AB$ is an odd number
- (D) B must be an identity matrix if A is invertible matrix

Space for Rough Work

5. The equation of plane passes through origin and which is equally inclined to the lines $\frac{x-1}{2} = \frac{y+1}{-2} = \frac{z+2}{-1}$ and $\frac{x-3}{8} = \frac{y-2}{1} = \frac{z-1}{-4}$ is/are -
- (A) $x + 5y - 2z = 0$ (B) $14x - 5y - 7z = 0$
 (C) $3x + 4y + 8z = 0$ (D) $2x + 7y - z = 0$
6. Let $A \equiv (1,2,3)$, $B \equiv (\alpha,4,6)$ and $C \equiv (1,2,\beta)$ are three vertices of ΔABC . If median AD is equally inclined with positive x,y and z axis, then which of the following is/are true
- (A) ordered pair (α,β) is $(3,2)$
 (B) coordinates of D are $(2,3,5)$
 (C) equation of median through B is $\frac{x-3}{4} = \frac{y-4}{4} = \frac{z-6}{7}$
 (D) equation of median through B is $\frac{x-3}{4} = \frac{y-4}{4} = \frac{z-6}{5}$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 16)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has Four options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

Paragraph for Questions 7 and 8

On the Argand plane consider $A(z_1)$, $B(z_2)$ and $C(z_3)$ as distinct complex numbers lying on the curve $|z| = \sqrt{3}$.

7. If a root of $z_1 z^2 + z_2 z + z_3 = 0$ has modulus unity then z_1, z_2, z_3 are in
 (A) A.P. (B) G.P. (C) H.P. (D) all of above
8. If $\triangle ABC$ is equilateral and $P(z_p)$ be any point such that $|z_1| = |z_p|$, then the value(s) of $|z_p - z_1|^2 + |z_p - z_2|^2 + |z_p - z_3|^2$ is
 (A) 9 (B) 6 (C) 18 (D) 3

Space for Rough Work

Paragraph for Questions 9 and 10

Let $P_1 : 2x - 3y + 6z + 8 = 0$ and $P_2 : 3x - 2y - 2z + 7 = 0$ be two planes and $A \left(0, \frac{29}{9}, \frac{5}{18} \right)$ lies on both planes. Two points $B(2, 0, -2)$ and C are such that line of intersection of the planes is the internal angle bisector of $\angle A$ of $\triangle ABC$ and $AB = AC$.

9. If co-ordinates of C are (α, β, γ) , then

- (A) $\alpha = -4$ (B) $\beta = 4$ (C) $\beta = -4$ (D) $\gamma = 2$

10. If the plane containing the point C and perpendicular to planes P_1 and P_2 is $ax + by + cz = 26$, then

- (A) $a = 18$ (B) $b = 11$ (C) $b = 22$ (D) $c = 13$

Space for Rough Work

SECTION-II : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Let $\vec{a}$, $\vec{b}$, $\vec{c}$ are unit vectors and $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = \cos \theta$; $\theta \in [0, \pi]$, then maximum value of θ is $k\pi$, where k is equal to
2. Given A and B are two non-singular matrices such that $B \neq I$, $A^5 = I$ and $AB^2 = BA$, then the least value of n for which $B^n = I$ is -
(Where $n \in \mathbb{N}$)
3. Let $A = \begin{bmatrix} \sqrt{3} & 1 & 0 \\ 1 & -\sqrt{3} & 0 \\ 0 & 0 & 2 \end{bmatrix}$ then the value of $\det(2A^T + AA^T + \text{adj}A)$ is equal to
4. Let w and z be complex numbers such that $|w| = 1$ and $|z| = 10$. Let $\theta = \arg\left(\frac{w-z}{z}\right)$. The minimum possible value of $\cot^2 \theta$ is

Space for Rough Work

5. Consider three urns Urn I contains 2 Red and 3 White balls, Urn II contains 2 Red and 4 White balls and Urn III contains 3 Red and 1 White ball. A Ball is drawn randomly from Urn I and placed unseen in Urn II. Now a ball is drawn randomly from Urn II and placed unseen in Urn III, then a ball is drawn from Urn III. If the probability that of the three successive draws, no two consecutive draws result in balls of same colour is p then $\frac{1}{p}$ is
6. PABC is a tetrahedron such that $\angle APB = \angle BPC = \angle CPA = 90^\circ$. If the sum of its six edges is constant S and let maximum volume of the tetrahedron be $\frac{5\sqrt{2}-7}{k} S^3$, then k is equal to
7. Let P be an interior point of $\triangle ABC$ such that $\vec{PA} + 2\vec{PB} + 3\vec{PC} = \vec{0}$, then ratio of area of $\triangle ABC$ to $\triangle APC$ is
8. A man is known to speak the truth 3 out of 4 times. He throws a die and reports that it is a six. The probability that it is actually a six is

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work